KIRAN SARASWATHY BHAVANAM RADHAKRISHNAN

Machine Learning Engineer | Python Developer | Applied Machine Learning

Kitchener, ON, Canada | Open to Remote, Hybrid, Onsite & Relocation | kiransr.careers@gmail.com

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PROFESSIONAL SUMMARY

Python-focused Software Developer with 1 year of professional experience and hands-on experience developing machine learning and deep learning applications. Built and deployed an end-to-end ML application achieving 87% accuracy and 91% recall, and contributed to a CNN-based tuberculosis detection system achieving 91.29% test accuracy. Skilled in Python, SQL, scikit-learn, XGBoost, TensorFlow, computer vision, model evaluation, and REST API integration. Holds a Master's in Computer Applications and is currently expanding into MLOps and applied AI.

TECHNICAL SKILLS

- Programming & Data: Python, SQL, Pandas, NumPy, SciPy, JavaScript
- Machine Learning & Deep Learning: scikit-learn, XGBoost, TensorFlow, Keras, CNNs, Feature Engineering, Cross-Validation, Model Evaluation, SHAP
- Computer Vision: OpenCV, Image Preprocessing, Segmentation, Data Augmentation, Transfer Learning
- ML Applications & Software Engineering: Streamlit, Flask, REST APIs, Git, Jupyter Notebook
- Web & Databases: React.js, Node.js, Express.js, MongoDB
- Currently Learning: MLOps, MLflow, LLMs, RAG, AI Agents

PROJECTS

Heart Disease Risk Prediction System | Live Demo | GitHub

Python | scikit-learn | XGBoost | SHAP | Streamlit

- Developed and deployed a reproducible end-to-end ML application, achieving 87% accuracy, 91% recall, and approximately 0.93 ROC-AUC.
- Implemented preprocessing, model training, stratified cross-validation, threshold evaluation, and SHAP-based explainability.
- Built an interactive Streamlit application for generating predictions and visualizing model results.

Tuberculosis Bacilli Detection Using Deep Learning | Academic Team Project

Python | TensorFlow | Keras | OpenCV | CNN | Image Segmentation

- Contributed to a custom six-convolutional-layer CNN for classifying tuberculosis bacilli in microscopy images, achieving 91.29% test accuracy.
- Applied contrast enhancement, masked SLIC segmentation, and augmentation of approximately 400 image patches into 4,000 training samples.
- Evaluated the model using training, validation, and test datasets, reporting 93.05% validation accuracy.

Real-Time Face Mask Detection | Computer Vision Application | GitHub

Python | TensorFlow | Keras | OpenCV | MobileNetV2 | Flask

- Built a MobileNetV2-based image classification system for detecting masked and unmasked faces.
- Integrated OpenCV webcam processing, deep learning inference, and a Flask application for real-time detection.

Banking AML Detection & MLOps Platform | In Progress

Python | Machine Learning | AML Transaction Analytics | MLOps (Planned)

- Analyzed the IBM synthetic AML dataset and scoped transaction-level money laundering risk classification.
- Designed a proposed pipeline for data validation, behavioral features, ML risk scoring, explainability, and future MLOps.

PROFESSIONAL EXPERIENCE

Software Developer | Tecforz Innovation | Kochi, India | Jul 2022 – Jun 2023

- Developed JavaScript features while modernizing a legacy application.
- Refactored and debugged code to improve maintainability and reliability.
- Integrated frontend components with REST APIs and MongoDB, collaborating on testing and feature delivery using Git.

CERTIFICATIONS & TRAINING

Google Data Analytics Professional Certificate | Structuring Machine Learning Projects - DeepLearning.AI

EDUCATION

- Post Graduate Diploma - Web Development | Conestoga College, Canada | 2024 | Distinction | GPA: 3.53/4.0
- Master of Computer Applications (MCA) | Cochin University of Science and Technology, India
- Bachelor of Science in Physics | St. Berchmans College, India